

CS 5330 HW1

Questions

1. Consider the situation where you want to put a camera in a car, looking towards the front of the car. A car is roughly 2.5m wide. You have a camera with a 1cm wide sensor with 2000 horizontal pixels.

- a. If you want the car to be at least 50 pixels wide at 100m, what does your focal length need to be? Give the answer in mm. Be sure to identify which equation is relevant to the question before starting.

First, we need to calculate the sensor distance. We know that the car appears as $10\text{mm} * 50/2000 = 0.25\text{mm}$ wide on the image sensor. Using the law of similar triangles, we can calculate sensor distance as $Z_{in} = 100,000\text{mm}/2,500\text{mm} * 0.25\text{mm} = 10\text{mm}$.

Next, with the sensor distance known, we can use the thin lens law to find the focal length: $1/f = 1/Z_{in} + 1/Z_{out} = 1/100,000\text{mm} + 1/10\text{mm} = 0.10001$. So $f = 9.999\text{mm}$

- b. How many pixels would a car be in the image at 12.5m with the same focal length?

Using thin lens law, calculate sensor distance: $Z_{in} = 1/(1/f - 1/Z_{out}) = 1/(1/9.999\text{mm} - 1/12,500\text{mm}) = 10.007\text{mm}$

Using the law of similar triangles, calculate image width: $= 2,500\text{mm} / 12,500\text{mm} * 10.007\text{mm} = 2.001\text{mm}$

Then calculate pixels: $px = 2000px * 2.001\text{mm} / 10\text{mm} = 400px$

- c. Given a 10mm lens, do you need to make any adjustments for the scene to be in focus from 20-100m? (Clarification: you might need to make an adjustment if the change is bigger than 1 in 100.) Identify which equation is relevant to the problem before answering.

Given a fixed focal length of 10mm, the law of thin lenses can be used to determine the relationship between changing Z_{out} from 20 - 100m.

First, for 20m, the necessary sensor position to keep perfect focus is: $Z_{in} = 1/(1/10\text{mm} - 1/20,000\text{mm}) = 10.005\text{mm}$

For 100m, the sensor position becomes: $Z_{in} = 1/(1/10\text{mm} - 1/100,000\text{mm}) = 10.001\text{mm}$

This is a difference of 0.004mm, less than 1 in 100 (0.01mm), therefore no adjustment is needed.

2. Do an internet search to find information about the response distributions for human rods and cones.

- a. What percentage of our cones are sensitive to long wavelength (red), medium wavelength (green), and short wavelength (blue)?

Red: 64%, Green: 32%, Blue: 2%

- b. When you look at the response curves for the red, green, and blue cones, how would you describe the differences between the three?

On a normalized cone response curve, the medium (green) and long (red) wavelength cone responses significantly overlap (especially in the green wavelength areas), while the short (blue) wavelength cone response is more separated. The short wavelength response also appears to have a sharper peak than the other two. Upon reading the website below, I also find that blue cones are not found in the fovea centralis.

- c. Given the information above, explain why the Bayer pattern on camera sensors has two green sensors for every one blue or red sensor.

Due to the fact that the medium and long wavelength cones in our eyes significantly overlap in the green wavelength sections, our eyes are more sensitive to green light. The Bayer pattern replicates the human eye and its increased green sensitivity with this pattern.

(Hint: a good website is <http://hyperphysics.phy-astr.gsu.edu/hbase/vision/rodcone.html>)

3. Do some research on Bayer Patterns / Bayer Filter (the wikipedia site is pretty good).

- a. What are some issues that arise because of the interpolation of colors to get RGB values at each pixel?

Since each pixel only can only sense one color, the other two colors for each pixel are “made up” interpolated values. Also, this can be problematic around edges, where interpolation might cause sharp changes in color of nearby pixels to show up as improbable colors.

- b. What is the benefit of saving RAW images and doing interpolation off the camera?

RAW data is the exact data that the camera sensors captured. Having access to the RAW image allows the user to have more freedom in what post-processing is done, for example, it allows the choice of choosing which de-mosaicing algorithm to use.

- c. Why would you want to use Cyan-Magenta-Yellow (CMY) instead of RGB filters? How would you get RGB values?

Since CMY make up the subtractive color system, the raw sensor values must be subtracted from white. Since RGB is additive, the sensor values are added up from black. While color can be captured with both methods, CMY sensors have improved light absorption compared to RBY and thus the overall image can have a better luminance.

4. Research the terms interleaved image representation and planar image representation. What is the difference between them?

Interleaved representation is where all the color values for each pixel are stored next together in computer memory. Planar image representation is where each color is grouped together.

5. You are watching the start of Raiders of the Lost Ark, and Indiana Jones is watching the boulder roll towards him. At time $t = 0$, he notes that the boulder appears to be 5 units across. At time $t = 5$, he notes that the boulder appears to be 10 units across. At what time will the boulder reach his current position (assuming he stayed there and the boulder is moving at a constant velocity)? What equation is appropriate to use for this question?

Using the law of similar triangles, we know that if the boulder has doubled in apparent size, then the distance to Indy must have halved. Therefore, assuming constant velocity, the boulder should reach his current position in another 5s, at $t = 10$ s.

6. Why is it important to clip the results of an exposure or brightness filter to the range $[0, \text{max}]$?

If the brightness is not clipped, over-exposure can be a problem, where the whites or glare in the image are too overpowering.